ON THE COMPUTABILITY OF EQUILIBRIUM STATES OF NON-UNIFORMLY EXPANDING MAPS.

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Abstract.

Contents

1. Introduction

1.1. Main results.

Theorem 1.1. *Let $(M, d, \mathcal{S})$ be a connected computable metric space, M be a recursively compact set. Then there exists an algorithm that satisfies the following property:*

For each local homeomorphism $f\colon M \to M$ and $\phi \in C^{0,\alpha}(M, d)$ satisfying Assumptions 1 and 2, this algorithm outputs a rational linear combination of finite dirac measures which are supported on some points in $\mathcal{S}$ as a 2^{-n}-approximation in the Wasserstein–Kantorovich metric W_d for the unique equilibrium state μ_ϕ for f and ϕ, after inputting the following data in this algorithm:

(i) *an algorithm computing the function ϕ,*
(ii) *an algorithm computing the map f,*
(iii) *a rational number Q such that $|\phi|_{\alpha,\sigma} \leqslant Q$, and*
(iv) *the integer n.*

2. Notation

The chordal metric σ on the Riemann sphere $\widehat{\mathbb{C}}$ is defined as follows: $\sigma(z, w) := \frac{2|z-w|}{\sqrt{1+|z|^2}\sqrt{1+|w|^2}}$ for all z, $w \in \mathbb{C}$, and $\sigma(\infty, z) = \sigma(z, \infty) := \frac{2}{\sqrt{1+|z|^2}}$ for all $z \in \mathbb{C}$. Let S^2 denote an oriented topological 2-sphere. We use $\mathbb{N}$ to denote the set of integers greater than or equal to 1 and $\mathbb{N}^* := \bigcup_{k\in\mathbb{N}} \mathbb{N}^k$. We write $\mathbb{N}_0 \coloneqq \{0\} \cup \mathbb{N}$ and $\mathbb{N}_0^* \coloneqq \{0\} \cup \mathbb{N}^*$. We denote by $\mathbb{Q}^+$ (resp. $\mathbb{R}^+$) the set of all positive rational (resp. real) numbers. The symbol log denotes the natural logarithm. For $x \in \mathbb{R}$, we define $\lfloor x \rfloor$ as the greatest integer $\leqslant x$, $\lceil x \rceil$ as the smallest integer $\geqslant x$, and $x^+ = (x)^+ \coloneqq \max\{x, 0\}$. For a function $f \colon X \to \mathbb{R}$ on a set X, we define $f^+ = (f)^+ \colon X \to \mathbb{R}$ by $f^+(x) \coloneqq (f(x))^+$ for each $x \in X$. The cardinality of a set A is denoted by $\operatorname{card} A$.

Consider a map $f \colon X \to X$ on a set X. We write f^n for the n-th iterate of f, and $f^{-n} \coloneqq (f^n)^{-1}$, for each $n \in \mathbb{N}$. We set $f^0 \coloneqq \mathrm{id}_X$, the identity map on X. For a real-valued function $\phi \colon X \to \mathbb{R}$, we write $S_n\phi(x) = S_n^f\phi(x) \coloneqq \sum_{m=0}^{n-1} \phi(f^m(x))$ for $x \in X$ and $n \in \mathbb{N}_0$. We omit the superscript f when the map f is clear from the context. When $n = 0$, by definition $S_0\phi = 0$.

Let (X, d) be a metric space. We denote by $\mathcal{B}(X)$ the σ-algebra of all Borel subsets of X. For each subset $Y \subseteq X$, we denote the diameter of Y by $\operatorname{diam}_d Y \coloneqq \sup\{d(x, y) : x, y \in Y\}$, the interior of Y by $\operatorname{int}_\circ Y$, and the characteristic function of Y by $\mathbb{1}_Y$.

For each $r \in \mathbb{R}$ and each $x \in X$, we denote the open (resp. closed) ball of radius r centered at x by $B_d(x, r) \coloneqq \{y \in X : d(x, y) < r\}$. For each $r \in \mathbb{R}$ and each nonempty set $K \subseteq X$, we define $d(x, K) \coloneqq \inf_{y \in K} d(x, y)$, and $B_d(K, r) \coloneqq \{x \in X : d(x, K) < r\}$. We often omit the metric d in the subscript when it is clear from the context.

For a compact metric space (X, d), we denote by $C(X)$ the space of continuous functions from X to $\mathbb{R}$, and by $\mathcal{M}(X)$ (resp. $\mathcal{P}(X)$) the set of finite signed Borel measures (resp. Borel probability measures) on X. Let $g \colon X \to X$ be a Borel-measurable transformation. We denote by $\mathcal{M}(X, g)$ the set of g-invariant Borel probability measures on X. Moreover, for each Borel subset $C \in \mathcal{B}(X)$, $\mathcal{P}(X; C)$ denotes the set $\{\mu \in \mathcal{P}(X) : \mu(C) = 1\}$. By the Riesz representation theorem, we can identify the dual of $C(X)$ with the space $\mathcal{M}(X)$. For $\mu \in \mathcal{M}(X)$, we use $\|\mu\|$ to denote the total variation norm of μ, $\operatorname{supp} \mu$ to denote the support of μ, and

$$\langle \mu, u \rangle \coloneqq \int u \, \mathrm{d}\mu$$

for each μ-integrable Borel function u on X. If we do not specify otherwise, we equip $C(X)$ with the uniform norm $\|\cdot\|_{C(X)} \coloneqq \|\cdot\|_\infty$, and equip $\mathcal{M}(X)$, $\mathcal{P}(X)$, and $\mathcal{M}(X, g)$ with the weak* topology.

The space of real-valued Hölder continuous functions with an exponent $\alpha \in (0, 1]$ on a compact metric space (X, d) is denoted as $C^{0,\alpha}(X, d)$. For each $\phi \in C^{0,\alpha}(X, d)$,

$$|\phi|_{\alpha,d} \coloneqq \sup\{|\phi(x) - \phi(y)|/d(x, y)^\alpha : x, y \in X, x \neq y\}. \tag{2.1}$$

For a complete separable metric space (X, d), we recall the Wasserstein–Kantorovich metric W_d on $\mathcal{P}(X)$ given by

$$W_d(\mu, \nu) \coloneqq \sup\bigl\{|\langle \mu, f \rangle - \langle \nu, f \rangle| : f \in C^{0,1}(X, d), |f|_{1,d} \leqslant 1\bigr\}. \tag{2.2}$$

Note that for Borel probability measures in $\mathcal{P}(X)$, the convergence in W_d is equivalent to the convergence in the weak* topology (see e.g. [Vi09, Corollary 6.13]).

3. Preliminaries

3.1. Computable analysis.

We recall fundamental notions and results from recursion theory and computable analysis.[1] We present, in order, definitions and results concerning the computability of real numbers, computable structures on metric spaces, computability of open sets, functions, compact sets, and probability measures.

Computability over the reals. We begin by reviewing basic notations and concepts from classical recursion theory; for an introduction, see e.g. [Bri94, Chapter 3].

Definition 3.1 (Effective enumeration and recursively enumerable set). Let $S \subseteq \mathbb{N}^*$ be a nonempty set. An *effective enumeration* of S is a sequence $\{x_i\}_{i \in \mathbb{N}}$ with $S = \{x_i : i \in \mathbb{N}\}$ such that there exists an algorithm that, for each $i \in \mathbb{N}$, upon input i, outputs x_i.

[1] Our notion of algorithm is consistent with *Type-2 machines* defined in [We00, Definition 2.1.1].

Moreover, a set $I \subseteq \mathbb{N}^*$ is said to be a *recursively enumerable set*[2] if $I = \emptyset$ or there exists an effective enumeration of I.

For brevity, the symbol I denotes a nonempty recursively enumerable set throughout this subsection. Note that $\mathbb{N}^k$, for $k \in \mathbb{N}$, and $\mathbb{N}^*$ are all recursively enumerable sets by Definition 3.1.

Definition 3.2 (Partial recursive and recursive function). Let $\{i_n\}_{n\in\mathbb{N}}$ be an effective enumeration of I. We say that $f\colon I \to \mathbb{N}_0^*$ is *partial recursive* if there exists an algorithm that, for each $n \in \mathbb{N}$, on input n, outputs $f(i_n)$ if $f(i_n) \in \mathbb{N}^*$, and runs forever otherwise, namely, if $f(i_n) = 0$. We say that $f\colon I \to \mathbb{N}_0^*$ is *recursive* if f is a partial recursive function with $f(I) \subseteq \mathbb{N}^*$.

We now define the computability of real numbers.

Definition 3.3 (Computable real number). A real number x is called *computable* if there exist three recursive functions $f\colon \mathbb{N} \to \mathbb{N}$, $g\colon \mathbb{N} \to \mathbb{N}$, and $h\colon \mathbb{N} \to \mathbb{N}$ such that $\left|(-1)^{h(n)} f(n)/g(n) - x\right| < 2^{-n}$ for all $i \in I$ and $n \in \mathbb{N}$.

Let $\{x_i\}_{i\in I}$ be a sequence of real numbers. We say that $\{x_i\}_{i\in I}$ is a *sequence of uniformly computable real numbers* if there exist three recursive functions $f\colon \mathbb{N} \times I \to \mathbb{N}$, $g\colon \mathbb{N} \times I \to \mathbb{N}$, and $h\colon \mathbb{N} \times I \to \mathbb{N}$ such that $\left|(-1)^{h(n,i)} f(n,i)/g(n,i) - x_i\right| < 2^{-n}$ for all $i \in I$ and $n \in \mathbb{N}$.

Clearly, $x \in \mathbb{R}$ is computable if and only if $\{x_i\}_{i\in\mathbb{N}}$ defined by $x_i := x$ for all $i \in \mathbb{N}$ is uniformly computable. For analogous concepts in the sequel, we will define the uniform sequence version and regard the individual case as the special case of constant sequences.

Computable metric spaces.

Definition 3.4 (Computable metric space). A *computable metric space* is a triple $(X, \rho, \mathcal{S})$ satisfying that

(i) (X, ρ) is a separable metric space,
(ii) $\mathcal{S} = \{s_n\}_{n\in\mathbb{N}}$ forms a countable dense subset $\{s_n : n \in \mathbb{N}\}$ of X, and
(iii) $\{\rho(s_m, s_n)\}_{(m,n)\in\mathbb{N}^2}$ is a sequence of uniformly computable real numbers.

The points in $\mathcal{S}$ are called *ideal*. Since $\mathbb{N}^3$ is recursively enumerable, the collection $\mathcal{B} := \{B(s_i, m/n) : i,\, m,\, n \in \mathbb{N}\}$ can be enumerated as $\{B_l\}_{l\in\mathbb{N}}$ satisfying the following: there exists an algorithm that, for each $l \in \mathbb{N}$, upon input l, outputs i, m, $n \in \mathbb{N}$ with $B_l = B(s_i, m/n)$. We call the elements in $\mathcal{B}$ *ideal balls* and such an enumeration of $\mathcal{B}$ an *effective enumeration of ideal balls* in $(X, \rho, \mathcal{S})$.

We then define the computability of points in a computable metric space.

Definition 3.5 (Computable point). Let $(X, \rho, \mathcal{S})$ be a computable metric space with $\mathcal{S} = \{s_i\}_{i\in\mathbb{N}}$, and $\{x_i\}_{i\in I}$ be a sequence of points in X. Then $\{x_i\}_{i\in I}$ is called *uniformly computable* (in $(X, \rho, \mathcal{S})$) if there exists a recursive function $f\colon \mathbb{N} \times I \to \mathbb{N}$ such that $\rho\big(s_{f(n,i)}, x_i\big) < 2^{-n}$ for all $n \in \mathbb{N}$ and $i \in I$. Moreover, a point x in X is *computable* (in $(X, \rho, \mathcal{S})$) if $\{x_i\}_{i\in\mathbb{N}}$ defined by $x_i := x$ for all $i \in \mathbb{N}$ is uniformly computable.

We now specify the computable structure on $\mathbb{R}$. Let $\mathcal{S}_{\mathbb{Q}} = \{q_n\}_{n\in\mathbb{N}}$ be the enumeration of $\mathbb{Q}$ induced by an effective enumeration of $\mathbb{N}^3$ via the mapping $(a, b, c) \mapsto (-1)^c a/b$. Note that $\{d_{\mathbb{R}}(q_m, q_n)\}_{(m,n)\in\mathbb{N}^2}$ is a sequence of uniformly computable real numbers, where $d_{\mathbb{R}}$ is the Euclidean metric. Then the triple $\big(\mathbb{R},\, d_{\mathbb{R}},\, \mathcal{S}_{\mathbb{Q}}\big)$ forms a computable metric space according to Definition 3.4. A similar construction provides a computable structure for $\mathbb{R}^+$. In this article, we fix these as the standard computability structures on $\mathbb{R}$ and $\mathbb{R}^+$. It is clear that under these structures, Definitions 3.3 and 3.5 are equivalent for the computability of real numbers. That is, a sequence of reals is uniformly computable in one sense if and only if it is in the other.

We also consider a weaker notion of computability over $\mathbb{R}$ that leverages its natural ordered structure.

[2]We emphasize that recursively enumerable sets in this article are subsets of $\mathbb{N}^*$.

Definition 3.6 (Semi-computable real number). Let $\{x_i\}_{i\in I}$ be a sequence of real numbers. We say that $\{x_i\}_{i\in I}$ is *uniformly lower* (resp. *upper*) *semi-computable* if there exist three recursive functions $f\colon \mathbb{N}\times I \to \mathbb{N}$, $g\colon \mathbb{N}\times I \to \mathbb{N}$, and $h\colon \mathbb{N}\times I \to \mathbb{N}$ such that for each $i \in I$, $\bigl\{(-1)^{h(n,i)} f(n,i)/g(n,i)\bigr\}_{n\in\mathbb{N}}$ is non-decreasing (resp. non-increasing) and converges to x_i as $n \to +\infty$. Moreover, a real number x is called *lower* (resp. *upper*) *semi-computable* if the sequence $\{x_i\}_{i\in\mathbb{N}}$ defined by $x_i \coloneqq x$ for each $i \in \mathbb{N}$ is uniformly lower (resp. upper) semi-computable.

Lower semi-computable open sets. We define an effective version of open sets and collect some relevant results.

Let $(X,\,\rho,\,\mathcal{S})$ be a computable metric space. Let $\mathcal{B}$ be the set of ideal balls, and $\{B_l\}_{l\in\mathbb{N}}$ be an effective enumeration of ideal balls in $(X,\,\rho,\,\mathcal{S})$. We define the set $\mathcal{B}_0 \coloneqq \mathcal{B}\cup\{\emptyset\}$ of *extended ideal balls* and an enumeration $\{D_l\}_{l\in\mathbb{N}}$ of $\mathcal{B}_0$ such that $D_1 = \emptyset$ and $D_l = B_{l-1}$ for each integer $l \geqslant 2$. We call such an enumeration an *effective enumeration of extended ideal balls* in $(X,\,\rho,\,\mathcal{S})$.

Definition 3.7 (Lower semi-computable open set). Let $(X,\,\rho,\,\mathcal{S})$ be a computable metric space, and $\{D_l\}_{l\in\mathbb{N}}$ be an effective enumeration of extended ideal balls. Then a sequence $\{U_i\}_{i\in I}$ of open sets in X is said to be *uniformly lower semi-computable open* (in $(X,\,\rho,\,\mathcal{S})$) if there exists a recursive function $f\colon \mathbb{N}\times I \to \mathbb{N}$ such that $U_i = \bigcup_{n\in\mathbb{N}} D_{f(n,i)}$ for each $i \in I$. Moreover, an open set $U \subseteq X$ is called *lower semi-computable open* (in $(X,\,\rho,\,\mathcal{S})$) if the sequence $\{U_i\}_{i\in\mathbb{N}}$ defined by $U_i \coloneqq U$ for $i \in \mathbb{N}$ is uniformly lower semi-computable open.

The above definition of a lower semi-computable open set differs slightly from the ones in [BBRY11, Definition 3.4] and [BRY14, Definition 2.4]. In our definition, we use extended ideal balls, which include the empty set $\emptyset$.

The term *recursively open set* in the literature (e.g. [GHR11, Subsection 2.2 and Definition 2.4] and [HR09, Subsection 3.3]) is equivalent to the notion of *lower semi-computable open set* defined above. A detailed discussion of this equivalence is provided in [He25, Subsection 3.3].

Note that we can algorithmically decide whether $s \in B$ for each ideal point $s \in \mathcal{S}$ and each extended ideal ball $B \in \mathcal{B}_0$. The following result then follows immediately from Definition 3.7 (see e.g. [He25, Proposition 3.9]).

Proposition 3.8. *Let $(X,\,\rho,\,\mathcal{S})$ be a computable metric space with $\mathcal{S} = \{s_n\}_{n\in\mathbb{N}}$. Assume that $\{U_i\}_{i\in I}$ is uniformly lower semi-computable open. Then there exists a recursively enumerable set $E \subseteq \mathbb{N}\times I$ such that $\{s_n : (n,i) \in E_i\} = \{s_n : n \in \mathbb{N}\} \cap U_i$, where $E_i \coloneqq \{(n,i) \in E : n \in \mathbb{N}\}$ for each $i \in I$.*

The following two results are two classical results in computable analysis which both follow immediately from Definitions 3.1 and 3.7 (see e.g. [He25, Propositions 3.10 & 3.11]).

Proposition 3.9. *Let $(X,\,\rho,\,\mathcal{S})$ be a computable metric space. Assume that H and L are two nonempty recursively enumerable sets with $L \subseteq I \times H$, and that $\{U_{i,h}\}_{(i,h)\in L}$ is uniformly lower semi-computable open. Then $\{\bigcup\{U_{i,h} : (i,h) \in L_h\}\}_{h\in H}$ is uniformly lower semi-computable open, where $L_h \coloneqq \{(i,h) \in L : i \in I\}$ for each $h \in H$. In particular, if $\{U_i\}_{i\in I}$ is uniformly lower semi-computable open, then $\bigcup_{i\in I} U_i$ is lower semi-computable open.*

Proposition 3.10. *Let $(X,\,\rho,\,\mathcal{S})$ be a computable metric space. Assume that $\{r_i\}_{i\in I}$ is a sequence of uniformly lower semi-computable real numbers and $\{x_i\}_{i\in I}$ is uniformly computable in $(X,\,\rho,\,\mathcal{S})$. Then $\{B(x_i, r_i)\}_{i\in I}$ is uniformly lower semi-computable open.*

Computability of functions. We begin with the definition of oracles for points.

Definition 3.11 (Oracle). Let $(X,\,\rho,\,\mathcal{S})$ be a computable metric space with $\mathcal{S} = \{s_i\}_{i\in\mathbb{N}}$, and $x \in X$. We say that a function $\tau\colon \mathbb{N} \to \mathbb{N}$ is an *oracle* for x if $\rho(s_{\tau(n)}, x) < 2^{-n}$ for each $n \in \mathbb{N}$.

With the above definition, computable functions can be defined as follows.

Definition 3.12 (Computable function). Let $(X, \rho, \mathcal{S})$ and $(X', \rho', \mathcal{S}')$ be computable metric spaces with $\mathcal{S} = \{s_n\}_{n\in\mathbb{N}}$ and $\mathcal{S}' = \{s'_n\}_{n\in\mathbb{N}}$. Assume that $\{i_n\}_{n\in\mathbb{N}}$ is an effective enumeration of I, and $C_i \subseteq X$ for each $i \in I$. Then a sequence $\{f_i\}_{i\in I}$ of functions $f_i\colon X \to X'$ is called a *sequence of uniformly computable functions with respect to* $\{C_i\}_{i\in I}$ if there exists an algorithm that, for all $l, n \in \mathbb{N}$, $x \in C_{i_n}$, and oracle τ for x, on input l, n, and τ, outputs $m \in \mathbb{N}$ with $\rho'(s'_m, f_{i_n}(x)) < 2^{-l}$. We often omit the phrase "with respect to $\{C_i\}_{i\in I}$" when $C_i = X$ for all $i \in I$. Moreover, a function $f\colon X \to X'$ is said to be a *computable function on* C if $\{f_i\}_{i\in\mathbb{N}}$, defined by $f_i := f$ for all $i \in \mathbb{N}$, is a sequence of uniformly computable functions with respect to $\{C_i\}_{i\in\mathbb{N}}$ defined by $C_i := C$ for all $i \in \mathbb{N}$. We often omit the phrase "with respect to C" when $C = X$.

Computable functions serve as an effective version of continuous functions. The following result provides examples of computable functions (see e.g. [We00, Examples 4.3.3 and 4.3.13.5]).

Example 3.13. The exponential function $\exp\colon \mathbb{R} \to \mathbb{R}$ and the logarithmic function $\log\colon \mathbb{R}^+ \to \mathbb{R}$ are computable functions.

We recall the following classical characterization of computable functions (cf. [RY21a, Proposition 5.2.14] and [BBRY11, Proposition 3.6]; see also [He25, Proposition 3.17]).

Proposition 3.14. *Let $(X, \rho, \mathcal{S})$ and $(X', \rho', \mathcal{S}')$ be computable metric spaces. Suppose $\{B'_n\}_{n\in\mathbb{N}}$ is an effective enumeration of ideal balls in $(X', \rho', \mathcal{S}')$. Given $f_i\colon X \to X'$ and $C_i \subseteq X$ for each $i \in I$, the following statements are equivalent:*

(i) *The sequence $\{f_i\}_{i\in I}$ is a sequence of uniformly computable functions with respect to $\{C_i\}_{i\in I}$.*

(ii) *There exists a sequence $\{U_{n,i}\}_{(n,i)\in\mathbb{N}\times I}$ of uniformly lower semi-computable open sets in $(X, \rho, \mathcal{S})$ such that $f_i^{-1}(B'_n) \cap C_i = U_{n,i} \cap C_i$ for all $i \in I$ and $n \in \mathbb{N}$.*

(iii) *For each nonempty recursively enumerable set M and each sequence $\{V'_m\}_{m\in M}$ of uniformly lower semi-computable open sets, there exists a sequence $\{W_{m,i}\}_{(m,i)\in M\times I}$ of uniformly lower semi-computable open sets in $(X, \rho, \mathcal{S})$ such that $f_i^{-1}(V'_m) \cap C_i = W_{m,i} \cap C_i$ for all $m \in M$ and $i \in I$.*

We now define a notion of weaker computability property for functions.

Definition 3.15 (Semi-computable function). Let $(X, \rho, \mathcal{S})$ be a computable metric space, $\{i_n\}_{n\in\mathbb{N}}$ be an effective enumeration of I, and $C_i \subseteq X$ for each $i \in I$. A sequence $\{f_i\}_{i\in I}$ of functions $f_i\colon X \to \mathbb{R}$ is a *sequence of uniformly upper* (resp. *lower*) *semi-computable functions with respect to* $\{C_i\}_{i\in I}$ if there exists an algorithm that, for all $l, n \in \mathbb{N}$, $x \in C_{i_n}$, and oracle τ for x, on input l, n, and τ, outputs $q_{l,n,\tau} \in \mathbb{Q}$ such that for each $n \in \mathbb{N}$, each $x \in C_{i_n}$, and each oracle τ for x, $\{q_{l,n,\tau}\}_{l\in\mathbb{N}}$ is non-increasing (resp. non-decreasing) and converges to $f_{i_n}(x)$ as $l \to +\infty$. We often omit the phrase "with respect to $\{C_i\}_{i\in I}$" when $C_i = X$ for each $i \in I$. Moreover, a function $f\colon X \to \mathbb{R}$ is said to be an *upper* (resp. a *lower*) *semi-computable function on* C if $\{f_i\}_{i\in\mathbb{N}}$ defined by $f_i := f$ for each $i \in \mathbb{N}$, is a sequence of uniformly upper (resp. lower) semi-computable functions with respect to $\{C_i\}_{i\in\mathbb{N}}$ defined by $C_i := C$ for all $i \in \mathbb{N}$. We often omit the phrase "with respect to C" when $C = X$.

The following proposition is an immediate consequence of Proposition 3.14 (see e.g. [He25, Proposition 3.19]).

Proposition 3.16. *Let $(X, \rho, \mathcal{S})$ be a computable metric space, and $\mathcal{S}_{\mathbb{Q}} = \{q_n\}_{n\in\mathbb{N}}$. Given $f_i\colon X \to \mathbb{R}$ and $C_i \subseteq X$ for all $i \in I$, the following statements are equivalent:*

(i) *The sequence $\{f_i\}_{i\in I}$ is a sequence of uniformly upper (resp. lower) semi-computable functions with respect to $\{C_i\}_{i\in I}$.*

(ii) *There exists a sequence $\{U_{n,i}\}_{(n,i)\in\mathbb{N}\times I}$ of uniformly lower semi-computable open sets in $(X, \rho, \mathcal{S})$ such that $f_i^{-1}(Q_n) \cap C_i = U_{n,i} \cap C_i$ with $Q_n := (-\infty, q_n)$ (resp. $Q_n := (q_n, +\infty)$) for all $i \in I$ and $n \in \mathbb{N}$.*

(iii) *For each nonempty recursively enumerable set L and each sequence $\{r_l\}_{l\in L}$ of uniformly computable real numbers, there exists a sequence $\{W_{l,i}\}_{(l,i)\in L\times I}$ of uniformly lower semi-computable open sets in $(X,\,\rho,\,\mathcal{S})$ such that $f_i^{-1}(R_l)\cap C_i = W_{l,i}\cap C_i$ with $R_l \coloneqq (-\infty, r_l)$ (resp. $R_l \coloneqq (r_l, +\infty)$) for all $l \in L$ and $i \in I$.*

The following result indicates that the characteristic function of a lower semi-computable open set is a lower semi-computable function (see e.g. [He25, Proposition 3.33]).

Proposition 3.17. *Let $(X,\,\rho,\,\mathcal{S})$ be a computable metric space. Assume that $\{U_i\}_{i\in I}$ is uniformly lower semi-computable open. Then there exists a sequence $\{h_{n,i}\}_{(n,i)\in\mathbb{N}\times I}$ of uniformly computable functions $h_{n,i}\colon X\to\mathbb{R}$ such that for each $i\in I$, the following properties are satisfied:*

(i) *For each $x\in X$, $\{h_{n,i}(x)\}_{n\in\mathbb{N}}$ is non-decreasing and converges to $\mathbb{1}_{U_i}(x)$ as $n\to+\infty$.*
(ii) *For each $n\in\mathbb{N}$, $h_{n,i}(x)\geqslant 0$ for each $x\in X$ and $h_{n,i}(x)=0$ for each $x\notin U_i$.*

Recursively compact sets and recursively precompact metric spaces. Here we recall the definitions of recursive compactness and recursive precompactness. For a more detailed discussion, see [GHR11, Section 2].

Definition 3.18 (Recursively compact set). Let $(X,\,\rho,\,\mathcal{S})$ be a computable metric space with $\mathcal{S}=\{s_i\}_{i\in\mathbb{N}}$, and $\{i_l\}_{l\in\mathbb{N}}$ be an effective enumeration of I. A sequence $\{K_i\}_{i\in I}$ of compact sets in X is called *uniformly recursively compact* (in $(X,\,\rho,\,\mathcal{S})$) if there exists an algorithm that, for each $n\in\mathbb{N}$, each sequence $\{m_n\}_{n=1}^p$ of integers, and each sequence $\{q_n\}_{n=1}^p$ of positive rational numbers, upon input, halts if and only if $K_{i_l}\subseteq\bigcup_{n=1}^p B(s_{m_n}, q_n)$. Moreover, a set $K\subseteq X$ is called *recursively compact* (in $(X,\,\rho,\,\mathcal{S})$) if the sequence $\{K_i\}_{i\in\mathbb{N}}$ defined by $K_i\coloneqq K$ for each $i\in\mathbb{N}$, is uniformly recursively compact.

Note that for each compact set K and each function $f\colon\mathbb{N}\to\mathbb{N}$, $K\subseteq\bigcup_{n\in\mathbb{N}}D_{f(n)}$ if and only if $K\subseteq\bigcup_{n=1}^k D_{f(n)}$ for some $k\in\mathbb{N}$. This implies the following result.

Proposition 3.19. *Let $(X,\,\rho,\,\mathcal{S})$ be a computable metric space. Suppose $\{h_m\}_{m\in\mathbb{N}}$ (resp. $\{l_n\}_{n\in\mathbb{N}}$) is an effective enumeration of a nonempty recursively enumerable set H (resp. L). Assume that $\{K_h\}_{h\in H}$ is uniformly recursively compact and $\{U_l\}_{l\in L}$ is uniformly lower semi-computable open. Then there exists an algorithm that, for all $m,\,n\in\mathbb{N}$, upon input, halts if and only if $K_{h_m}\subseteq U_{l_n}$.*

We collect some fundamental properties of recursively compact sets (cf. [GHR11, Propositions 1 & 3]; see also [He25, Proposition 3.23]).

Proposition 3.20. *Let $(X,\,\rho,\,\mathcal{S})$ be a computable metric space. Assume that X is recursively compact, and $\{K_i\}_{i\in I}$ is uniformly recursively compact. Then the following statements are true:*

(i) *Let $x_i\in X$ for each $i\in I$. Then $\{x_i\}_{i\in I}$ is uniformly computable if and only if the sequence $\{\{x_i\}\}_{i\in I}$ of singletons is uniformly recursively compact.*
(ii) *$\{X\smallsetminus K_i\}_{i\in I}$ is uniformly lower semi-computable open.*
(iii) *If $\{U_i\}_{i\in I}$ is uniformly lower semi-computable open, then $\{K_i\smallsetminus U_i\}_{i\in I}$ is uniformly recursively compact.*
(iv) *If $\{f_i\}_{i\in I}$ is a sequence of uniformly lower (resp. upper) semi-computable functions $f_i\colon X\to\mathbb{R}$ with respect to $\{K_i\}_{i\in I}$, then $\{\inf_{x\in K_i} f_i(x)\}_{i\in I}$ (resp. $\{\sup_{x\in K_i} f_i(x)\}_{i\in I}$) is uniformly lower (resp. upper) semi-computable.*
(v) *If $\{T_i\}_{i\in I}$ is a sequence of uniformly computable functions $T_i\colon X\to X$ with respect to $\{K_i\}_{i\in I}$, then $\{T_i(K_i)\}_{i\in I}$ is uniformly recursively compact.*

Next, we investigate whether the property of uniform computability for recursively compact sets is preserved under the union and intersection.

Proposition 3.21. *Let $(X,\,\rho,\,\mathcal{S})$ be a computable metric space. Suppose X is recursively compact, H and L are two nonempty recursively enumerable sets with $L \subseteq I \times H$, and $\{K_{i,h}\}_{(i,h)\in L}$ is uniformly recursively compact. Denote $L_h \coloneqq \{(i,h) \in L : i \in I\}$ for each $h \in H$. Then the following statements are true:*

(i) $\{\bigcap\{K_{i,h} : (i,h) \in L_h\}\}_{h\in H}$ *is uniformly recursively compact.*

(ii) *If the function $F\colon H \to \mathbb{N}$ defined by $F(h) \coloneqq \operatorname{card} L_h$ for $h \in H$ is recursive, then $\{\bigcup\{K_{i,h} : (i,h) \in L_h\}\}_{h\in H}$ is uniformly recursively compact.*

Proposition 3.21 (i) follows immediately from Proposition 3.9 and Proposition 3.20 (ii) and (iii). Moreover, Proposition 3.21 (ii) follows from Definition 3.18. As a corollary of Proposition 3.21 (ii), we obtain the following result, which is important in the proof of Theorem **??**.

Corollary 3.22. *Let $(X,\,\rho,\,\mathcal{S})$ be a computable metric space. Assume that X is recursively compact, $T\colon X \to X$ is a computable function, and $\{U_i\}_{i\in I}$ is uniformly lower semi-computable open. Then $\{V_{n,i}\}_{(n,i)\in\mathbb{N}\times I}$ is uniformly lower semi-computable open, where $V_{n,i}$ is defined inductively by setting $V_{1,i} \coloneqq U_i$ and $V_{n+1,i} \coloneqq T^{-1}(V_{n,i}) \cap U_i$ for each $n \in \mathbb{N}$ and each $i \in I$.*

Proof. Since T is a computable function, by Definition 3.2, we obtain that $\{T^n\}_{n\in\mathbb{N}_0}$ is a sequence of uniformly computable functions. Then by Proposition 3.14, $\{T^{-n}(U_i)\}_{(n,i)\in\mathbb{N}_0\times I}$ is a sequence of uniformly lower semi-computable open sets. Hence, since X is recursively compact, by Proposition 3.20 (iii), $\{X \smallsetminus T^{-n}(U_i)\}_{(n,i)\in\mathbb{N}_0\times I}$ is a sequence of uniformly recursively compact sets.

Define $L \subseteq \mathbb{N}_0 \times \mathbb{N} \times I$ by $L \coloneqq \{(m,n,i) \in \mathbb{N}_0 \times \mathbb{N} \times I : m < n\}$. Then L is a recursively enumerable set and $\{X \smallsetminus T^{-m}(U_i)\}_{(m,n,i)\in L}$ is uniformly recursively compact. Note that $\operatorname{card}\{m \in \mathbb{N}_0 : (m,n,i) \in L\} = n$ for all $n \in \mathbb{N}$ and $i \in I$. Then the function $F\colon \mathbb{N} \times I \to \mathbb{N}_0$ given by $F(n,i) \coloneqq \operatorname{card}\{m \in \mathbb{N}_0 : (m,n,i) \in L\}$ is a recursive function. Thus, by Proposition 3.21 (ii), we obtain that $\{\bigcup\{X \smallsetminus T^{-m}(U_i) : m \in \mathbb{N}_0,\, (m,n,i) \in L\}\}_{(n,i)\in\mathbb{N}\times I}$ is a sequence of uniformly recursively compact sets. Hence, since X is a recursively compact set, by Proposition 3.20 (ii), $\{X \smallsetminus \bigcup\{X \smallsetminus T^{-m}(U_i) : m \in \mathbb{N}_0,\, (m,n,i) \in L\}\}_{(n,i)\in\mathbb{N}\times I} = \{\bigcap\{T^{-m}(U_i) : m \in \mathbb{N}_0,\, (m,n,i) \in L\}\}_{(n,i)\in\mathbb{N}\times I}$ is uniformly lower semi-computable open. Since $V_{1,i} = U_i$ and $V_{n+1,i} = T^{-1}(V_{n,i}) \cap U_i$ for all $n \in \mathbb{N}$ and $i \in I$, it follows by induction that $V_{n+1,i} = \bigcap_{k=0}^{n} T^{-k}(U_i)$ for each $n \in \mathbb{N}_0$. Therefore, $\{V_{n,i}\}_{(n,i)\in\mathbb{N}\times I}$ is uniformly lower semi-computable open. $\square$

Moreover, given the recursive compactness of X, the computability of functions is preserved under a finite number of operations among additions and multiplications. We summarize this property in the following result (cf. [We00, Corollary 4.3.4]; see also [He25, Proposition 3.26]).

Proposition 3.23. *Let $(X,\,\rho,\,\mathcal{S})$ be a computable metric space in which X is recursively compact, and H be a nonempty recursively enumerable set. Assume that $\{f_i\}_{i\in I}$ (resp. $\{g_h\}_{h\in H}$) is a sequence of uniformly computable functions $f_i\colon X \to \mathbb{R}$ (resp. $g_h\colon X \to \mathbb{R}$). Then $\{f_i + g_h\}_{(i,h)\in I\times H}$, $\{f_i \cdot g_h\}_{(i,h)\in I\times H}$ are two sequences of uniformly computable functions.*

Next, we recall the definition of recursively precompact metric space.

Definition 3.24 (Recursively precompact metric space). Let $(X,\,\rho,\,\mathcal{S})$ be a computable metric space with $\mathcal{S} = \{s_i\}_{i\in\mathbb{N}}$. Then $(X,\,\rho,\,\mathcal{S})$ is called *recursively precompact* if there exists an algorithm that, for each $n \in \mathbb{N}$, on input n, outputs a finite subset $\{r_i : 1 \leqslant i \leqslant m\}$ of $\mathbb{N}$ such that $X = \bigcup_{i=1}^{m} B(s_{r_i}, 2^{-n})$.

Finally, we record the following useful characterization of complete recursively precompact metric spaces (see e.g. [GHR11, Proposition 4]).

Proposition 3.25. *Let $(X,\,\rho,\,\mathcal{S})$ be a computable metric space. Then X is recursively compact if and only if (X,ρ) is complete and $(X,\,\rho,\,\mathcal{S})$ is recursively precompact.*

Computability of probability measures. Building upon the theory of computable functions and recursively compact sets, we now discuss the computability of probability measures. We begin by reviewing the computable structure on the measure space $\mathcal{P}(X)$ introduced in [HR09, Section 4] (cf. [HR09, Proposition 4.1.3]; see also [He25, Proposition 3.29]).

Proposition 3.26. *Let $(X,\,\rho,\,\mathcal{S})$ be a computable metric space with $\mathcal{S}=\{s_n\}_{n\in\mathbb{N}}$. Assume that X is recursively compact in $(X,\,\rho,\,\mathcal{S})$. Then the following statements are true:*

(i) *There exists an enumeration $\mathcal{Q}_{\mathcal{S}}=\{\nu_k\}_{k\in\mathbb{N}}$ of the set of Borel probability measures that are supported on finitely many points in $\{s_n : n\in\mathbb{N}\}$ and assign rational values to them such that there exists an algorithm that, for each $k\in\mathbb{N}$, upon input k, outputs a sequence $\{n_l\}_{l=1}^{p}$ of integers and a sequence $\{q_l\}_{l=1}^{p}$ of positive rational numbers satisfying that $\sum_{l=1}^{p} q_l = 1$ and $\nu_k=\sum_{l=1}^{p} q_l\delta_{s_{n_l}}$.*

(ii) *$(\mathcal{P}(X),\,W_\rho,\,\mathcal{Q}_{\mathcal{S}})$ is also a computable metric space in which $\mathcal{P}(X)$ is recursively compact, where W_ρ is the Wasserstein–Kantorovich metric on $\mathcal{P}(X)$ (see (2.2)).*

Let $(X,\,\rho,\,\mathcal{S})$ be a computable metric space and assume that X is recursively compact. We endow the measure space $\mathcal{P}(X)$ with the computable structure $(\mathcal{P}(X),\,W_\rho,\,\mathcal{Q}_{\mathcal{S}})$ given by Proposition 3.26.

The computability of measures is then defined via Definition 3.5. Specifically, a sequence $\{\mu_i\}_{i\in I}$ of measures in $\mathcal{P}(X)$ is *a sequence of uniformly computable measures* if it is uniformly computable in $(\mathcal{P}(X),\,W_\rho,\,\mathcal{Q}_{\mathcal{S}})$, and a single measure $\mu\in\mathcal{P}(X)$ is *a computable measure* if the corresponding constant sequence consisting of μ is uniformly computable.

We now recall a key result on the computability of the integration function (cf. [HR09, Corollary 4.3.2]; see also [He25, Proposition 3.30]).

Proposition 3.27. *Let $(X,\,\rho,\,\mathcal{S})$ be a computable metric space. Assume that X is recursively compact in $(X,\,\rho,\,\mathcal{S})$, and that $\{f_i\}_{i\in I}$ is a sequence of uniformly computable functions $f_i\colon X\to\mathbb{R}$. Then the sequence $\{\mathcal{I}_i\}_{i\in I}$ of functions $\mathcal{I}_i\colon\mathcal{P}(X)\to\mathbb{R}$ defined by $\mathcal{I}_i(\mu)\coloneqq\langle\mu,f_i\rangle$ for $\mu\in\mathcal{P}(X)$ is a sequence of uniformly computable functions.*

As immediate corollaries of Proposition 3.27, we have the following results.

Corollary 3.28. *Let $(X,\,\rho,\,\mathcal{S})$ be a computable metric space. Assume that X is recursively compact in $(X,\,\rho,\,\mathcal{S})$, and that $\{f_i\}_{i\in I}$ is a sequence of uniformly upper (resp. lower) semi-computable functions $f_i\colon X\to\mathbb{R}$. Then the sequence $\{\mathcal{I}_i\}_{i\in I}$ of functions $\mathcal{I}_i\colon\mathcal{P}(X)\to\mathbb{R}$ given by $\mathcal{I}_i(\mu)\coloneqq\langle\mu,f_i\rangle$ is a sequence of uniformly upper (resp. lower) semi-computable functions.*

Corollary 3.29. *Let $(X,\,\rho,\,\mathcal{S})$ be a computable metric space in which X is recursively compact. Assume that H and L are two nonempty recursively enumerable sets with $L\subseteq I\times H$, $\{U_{i,h}\}_{(i,h)\in L}$ is uniformly lower semi-computable open in $(X,\,\rho,\,\mathcal{S})$, and $\{r_{i,h}\}_{(i,h)\in L}$ is a sequence of uniformly computable real numbers. Define, for each $i\in I$, $L_i\coloneqq\{(i,h)\in L : h\in H\}$ and $\mathcal{K}_i\coloneqq\{\mu\in\mathcal{P}(X) : \mu(U_{i,h})\leqslant r_{i,h}$ for each $(i,h)\in L_i\}$. Then $\{\mathcal{K}_i\}_{i\in I}$ is uniformly recursively compact in $(\mathcal{P}(X),\,W_\rho,\,\mathcal{Q}_{\mathcal{S}})$.*

Recall the definition of $\mathcal{M}(X,T;Y)$ from (**??**). The following result indicates the recursive compactness of the set $\mathcal{M}(X,T;Y)$ (cf. [BHLZ25, Lemma 4.12]; see also [He25, Proposition 3.36]).

Proposition 3.30. *Let $(X,\,\rho,\,\mathcal{S})$ be a computable metric space in which X is recursively compact, and $\{U_i\}_{i\in I}$ is a sequence of uniformly lower semi-computable open sets. Assume that $\{T_i\}_{i\in I}$ is a sequence of uniformly computable functions $T_i\colon X\to X$ with respect to $\{U_i\}_{i\in I}$. Then $\{\mathcal{M}(X,T_i;U_i)\}_{i\in I}$ is uniformly recursively compact in $(\mathcal{P}(X),\,W_\rho,\,\mathcal{Q}_{\mathcal{S}})$. In particular, if $\{T_i\}_{i\in I}$ is a sequence of uniformly computable functions, then $\{\mathcal{M}(X,T_i)\}_{i\in I}$ is uniformly recursively compact.*

The following result is useful in the proof of the main result of this article (see e.g. [He25, Proposition 3.35]).

Proposition 3.31. *Let $(X,\,\rho,\,\mathcal{S})$ be a computable metric space, and X be a recursively compact set in $(X,\,\rho,\,\mathcal{S})$. Then there exists a sequence $\{\tau_n\}_{n\in\mathbb{N}}$ of uniformly computable functions $\tau_n\colon X\to\mathbb{R}$ such that $\{\tau_n : n\in\mathbb{N}\}$ is dense in $C(X)$. Moreover, for all $\mu,\,\nu\in\mathcal{M}(X)$, $\mu(A)\geqslant\nu(A)$ for each $A\in\mathcal{B}(X)$ if and only if $\langle\mu,\tau_n^+\rangle\geqslant\langle\nu,\tau_n^+\rangle$ for each $n\in\mathbb{N}$.*

3.2. **Thermodynamic formalism.** We review basic concepts from ergodic theory. For more detailed discussions, we refer the reader to [Wa82, Section 4].

Let $(X,\mathcal{B},\mu)$ be a probability space. A *partition* $\xi=\{A_h : h\in H\}$ of $(X,\mathcal{B},\mu)$ is a disjoint collection of elements of $\mathcal{B}$ whose union is X, where H is a countable index set. For each pair of partitions $\xi=\{A_h : h\in H\}$ and $\eta=\{B_l : l\in L\}$ of X, their *join* is the partition $\xi\vee\eta\coloneqq\{A_h\cap B_l : h\in H,\,l\in L\}$.

Assume that $T\colon X\to X$ is a measure-preserving transformation of $(X,\mathcal{B},\mu)$. Consider a partition $\xi=\{A_h : h\in H\}$ of X. For each $n\in\mathbb{N}$, $T^{-n}(\xi)$ denotes the partition $\bigl\{T^{-1}(A_h) : h\in H\bigr\}$, and ξ_T^n denotes the join $\xi\vee T^{-1}(\xi)\vee\cdots\vee T^{-(n-1)}(\xi)$. The *entropy* of ξ is $H_\mu(\xi)\coloneqq-\sum_{h\in H}\mu(A_h)\log(\mu(A_h))\in[0,+\infty]$, where $0\log 0$ is defined to be zero. One can show that if $H_\mu(\xi)<+\infty$, then $\lim\limits_{n\to+\infty}H_\mu(\xi_T^n)/n$ exists (see e.g. [Wa82, Chapter 4]). We denote this limit by $h_\mu(T,\xi)$ and call it the *measure-theoretic entropy of T relative to ξ*. The *measure-theoretic entropy* of T for μ is defined as

$$h_\mu(T)\coloneqq\sup\{h_\mu(T,\xi) : \xi \text{ is a partition of } X \text{ with } H_\mu(\xi)<+\infty\}. \tag{3.1}$$

We now introduce thermodynamic formalism, a particular branch of ergodic theory. The main objects of study are the topological pressure and equilibrium states (see e.g. [PU10, Wa82]; for the general Borel-measurable setting used in Approach II, see e.g. [IT10, Definition 1.1], [DeT17, Section 2.3], and [DoT23, Chapter 1.4]).

Let (X,ρ) be a compact metric space, $T\colon X\to X$ be a Borel-measurable transformation such that $\mathcal{M}(X,T)\neq\emptyset$, and $\phi\colon X\to[-\infty,+\infty]$ be a Borel function. Then the *topological pressure* of the potential ϕ with respect to the transformation T is given by

$$P(T,\phi)\coloneqq\sup\bigl\{h_\mu(T)+\langle\mu,\phi\rangle : \mu\in\mathcal{M}(X,T) \text{ and } \langle\mu,\phi\rangle>-\infty\bigr\}. \tag{3.2}$$

A measure $\mu\in\mathcal{M}(X,T)$ that attains the supremum in (3.2) is called an *equilibrium state* for the transformation T and the potential ϕ. Denote the set of all such measures by $\mathcal{E}(T,\phi)$. In particular, when the potential ϕ is the constant function 0, we denote $h_{\mathrm{top}}(T)\coloneqq P(T,0)$ and say that a measure $\mu\in\mathcal{M}(X,T)$ is a *measure of maximal entropy* of T if $\mu\in\mathcal{E}(T,0)$.

4. Proof of Main Results

We study a class of non-uniformly expanding maps considered by Castro and Varandas in [?]. Let (M,d) be a compact connected separable metric space and $f\colon M\to M$ be a local homeomorphism. Let $\deg(f)$ denote the degree of f.

A list of assumptions that are applied in the rest of this paper is displayed below:

Assumption 1.

(A1) There exists a pair of computable functions $L\colon M\to\mathbb{R}^+$ and $r\colon M\to\mathbb{R}^+$ such that, for each $x\in M$, $f_x\coloneqq f|_{B(x,r(x))}\colon B(x,r(x))\to f(B(x,r(x)))$, the restriction of f to $B(x,r(x))$, is invertible and satisfies that

$$d(f_x^{-1}(y),f_x^{-1}(z))\leqslant L(x)\,d(y,z)\quad\text{for each pair of } y,\,z\in f(B(x,r(x))).$$

Moreover, there exist constants $\kappa>0$, $C_0\in\mathbb{R}$ and $\alpha_0>0$ so that $L(x)\geqslant\kappa$ and L is Hölder continous with constantes C_0 and α_0.

(A2) There exists a pair of constants $\sigma>1$ and $L\geqslant 1$, and an open subset $\mathcal{A}\subseteq M$ satisfying that $L(x)\leqslant L$ for each $x\in\mathcal{A}$ and $L(x)<\sigma^{-1}$ for each $x\notin\mathcal{A}$.

(A3) There exists a finite covering $\mathcal{U}=\{U_i\}_{i=1}^n$ of M by open subsets on which f is injective such that there exists an integer $q<\deg(f)$ satisfying that the subcover $\mathcal{U}'=\{U_i\}_{i=1}^q$ covers $\mathcal{A}$.

(A4) $\phi \in C^{0,\alpha}(M,d)$ satisfies the properties that $\|\phi\|_\infty < \varepsilon_\phi$ and $|e^\phi|_\alpha < \varepsilon_\phi e^{\inf_{x\in M}\phi(x)}$.

From Assumptions (A1) to (A3), we have that the map f is uniformly expanding in $\mathcal{A}^c$ while in $\mathcal{A}$ it is allowed some *controlled* contraction. Moreover, Assumption (A3) ensures that every point has at least one preimage in the uniformly expanding region $\mathcal{A}^c$.

We make the conditions on the constants of Assumption 1 sharper.

Assumption 2.

(B1) There exists a pair of constants $c > 0$ and $0 < \gamma < 1$ satisfying that $\sigma^{-(1-\gamma)}L^\gamma < e^{-2c} < 1$.

(B2) $e^{\varepsilon_\phi}\left(\dfrac{(\deg(f)-q)\sigma^{-\alpha} + qL^\alpha[1+(L-1)^\alpha]}{\deg(f)}\right) < 1.$

We recall ... in our context here.

Theorem 4.1. *Let (M,d) be a compact connected separable metric space, $f\colon M \to M$ be a local homeomorphism, and $\phi : M \to \mathbb{R}$ be a continuous function. Assume that f and ϕ satisfy Assumptions 1 and 2. Then there exists a unique equilibrium state μ for f with respect to ϕ.*

5. Preliminaries

5.1. **Hyperbolic times.** The concept of hyperbolic times was introduced in [**?**] and since then it has been greatly explored to maps whose lack of hyperbolicity is controlled.

The definition of [**?**] was given in a smooth context, here we use the generalized concept introduced in [**?**]. We say that n is a *hyperbolic time* for x if

$$\prod_{j=n-k}^{n-1} \|L(f^j(x))\| \leqslant e^{-ck}, \quad \text{for all } 1 \leqslant k < n.$$

Assumption 1 of non-uniform expansion is enough to guarantee the existence of infinitely many hyperbolic times for the points outside $\mathcal{A}$.

The key property of a hyperbolic time is that iterates of a map at hyperbolic times behave locally as uniformly expanding maps.

Add some reference lemma 2.7 and Prop 2.8

Lemma 5.1. *There exist $K, \delta_1 > 0$ such that for every $0 < \varepsilon \leqslant \delta$ and every hyperbolic time n for $x \in M$ the dynamic ball $B_\varepsilon(x,n)$ is mapped diffeomorphically under f^n onto the ball $B(f^n(x),\varepsilon)$. Moreover, for all Hölder continuous potential ϕ and all $y,z \in B_\varepsilon(x,n)$ we have*

(i) *$d(f^{n-j}(y), f^{n-j}(z)) \leqslant e^{-cj/4} d(f^n(y), f^n(z))$ for each $1 \leqslant j \leqslant n$;*

(ii) *$\dfrac{1}{K} \leqslant e^{S_n\phi(y) - S_n\phi(z)} \leqslant K$.*

5.2. **Cones and projective metrics.** This section is devoted to state some results regarding projective metrics, a notion that was first introduced by Birkhoff in [**?**] and has been used in a fruitful way to obtain spectral properties for the transfer operator. The reader can consult the monograph [**?**] for a more extended presentation on the subject.

Let E be a vector space over $\mathbb{R}$. A subset $\mathcal{C}$ of $E - \{0\}$ is called a *cone* in E if $C \cap (-C) = \{0\}$ and it satisfies:

$$u \in C \quad \text{and} \quad \lambda > 0 \quad \Rightarrow \quad \lambda \cdot u \in C.$$

A cone C is called *convex* if

$$u, v \in C \quad \text{and} \quad \lambda, \eta > 0 \quad \Rightarrow \quad \lambda \cdot u + \eta \cdot v \in C.$$

The closure of C, denoted by $\bar{C}$, is defined by

$$\bar{C} := \{v \in E\colon \text{ there are } u \in C \text{ and } \lambda_n \to 0 \text{ such that } (v + \lambda_n u) \in C \text{ for all } n \geqslant 1\}.$$

A cone $\mathcal{C}$ is called *closed* if $\bar{\mathcal{C}} = \mathcal{C} \cup \{0\}$.

Let $\mathcal{C}$ be a closed convex cone and given $u, v \in \mathcal{C}$ define

$$A(u,v) = \sup\{t > 0 \colon v - tu \in \mathcal{C}\} \text{ and } B(u,v) = \inf\{s > 0 \colon su - v \in \mathcal{C}\},$$

with the convention $\sup \emptyset = 0$ and $\inf \emptyset = +\infty$, where $\emptyset$ denotes the empty set.

We have that $A(u,v)$ is finite, $B(u,v)$ is positive and $A(u,v) \leqslant B(u,v)$ for all $u, v \in \mathcal{C}$ (see [?]). Define the projective metric by

$$\Theta(u,v) = \log\left(\frac{B(u,v)}{A(u,v)}\right),$$

with Θ possibly infinity in the case $A = 0$ or $B = +\infty$.

Note that $\Theta(u,v)$ is well-defined and takes values in $[0, +\infty]$. Since $\Theta(u,v) = 0$ if and only if $u = tv$ for some $t > 0$, Θ defines a pseudo-metric on $\mathcal{C}$. In this way, Θ induces a metric on a projective quotient space of $\mathcal{C}$ called the *projective metric of* $\mathcal{C}$.

Let E_1, E_2 be vector spaces over $\mathbb{R}$, if $L \colon E_1 \to E_2$ is a linear operator, and $\mathcal{C}_1, \mathcal{C}_2$ are convex cones in E_1, E_2, respectively, such that $L(\mathcal{C}_1) \subset \mathcal{C}_2$, then $\Theta_2(L(u), L(v)) \leqslant \Theta_1(u,v)$ for all $u, v \in \mathcal{C}_1$, where Θ_1 and Θ_2 are the projective metrics in $\mathcal{C}_1$ and $\mathcal{C}_2$, respectively.

In general, L need not be a strict contraction, that will be the case for instance if $L(\mathcal{C}_1)$ had finite diameter in $\mathcal{C}_2$ according to the next result.

Theorem 5.2. *Let $\mathcal{C}_1$ and $\mathcal{C}_2$ be closed convex cones in the Banach spaces E_1 and E_2, respectively. If $L \colon E_1 \to E_2$ is a linear operator such that $L(\mathcal{C}_1) \subset \mathcal{C}_2$ and $\Delta = \operatorname{diam}_{\Theta_2}(L(\mathcal{C}_1)) < \infty$, then*

$$\Theta_2\left(L(u), L(v)\right) \leqslant (1 - e^{-\Delta}) \cdot \Theta_1(u,v) \quad \textit{for all } u, v \in \mathcal{C}_1.$$

Note that, a linear bounded operator that is a strict contraction admits a fixed point. That is the key point to prove that the transfer operator admit a spectral gap in the space of Hölder continuous observables.

5.3. **The Ruelle-Perron-Frobenius operator on the space of Hölder continuous observables.** In this section we define the Ruelle-Perron-Frobenius operator also called transfer operator and state its nice spectral properties. The main goal is to show that there is a sequence of iterates of the transfer operator that converges exponentially to its eigenfunction with explicit constants.

The Ruelle-Perron-Frobenius operator $\mathcal{L}$ associated to $f : M \to M$ and $\phi : M \to \mathbb{R}$ is the linear operator defined by

$$\mathcal{L}(u)(x) = \sum_{y \in f^{-1}(x)} u(y) \exp(\phi(y)) \quad \text{for } u \in C(M) \text{ and } x \in M.$$

Note that $\mathcal{L}$ is a positive and bounded linear operator. Consider the resolvent set

$$Res(\mathcal{L}) := \{z \in \mathbb{C} \colon (zI - \mathcal{L}) \text{ has a bounded inverse}\},$$

and its complement set, called the spectrum set,

$$Spec(\mathcal{L}) := \{z \in \mathbb{C} \colon (zI - \mathcal{L}) \text{ has no bounded inverse}\}.$$

We say that the transfer operator $\mathcal{L}$ satisfies the *spectral gap property* if its spectrum set $Spec(\mathcal{L}) \subset \mathbb{C}$ admits a decomposition as follows: $Spec(\mathcal{L}) = \{\lambda\} \cup \Sigma$ where $\lambda \in \mathbb{R}$ is an eigenvalue for $\mathcal{L}$ associated to a one-dimensional eigenspace and Σ is strictly contained in a ball centered at zero and of radius strictly less than λ.

In thermodynamical formalism a problem of proving uniqueness of equilibrium states can be reduced to find an appropriate space on which the transfer operator admits a spectral gap property. The choice of this space depend in general on the underlying dynamics and on the class of potentials. In this setting a suitable space is the space of Hölder continuous observables.

Theorem 5.3 (Theorem A,[?]). *Let $f : M \to M$ be a local homeomorphism and $\phi : M \to \mathbb{R}$ be a potential satisfying Assumptions 1 and 2. Then the Ruelle-Perron-Frobenius operator has a spectral gap property in the space of Hölder continuous observables.*

The spectral gap property implies the existence of an eigenfunction for the dual of the transfer operator. This eigenfunction arises as the limit of a sequence of normalized transfer operators. This result was established by Castro and Varandas in [?]; however, their proof does not yield the quantitative estimates necessary for our purposes, particularly in establishing the computability of equilibrium states. For this reason, we will provide a complete proof of the theorem. The main idea is to prove that the transfer operator is a contraction in an appropriate cone of Holder continuous functions. Let us start with some definitions and techinical results.

Given $\delta > 0$ we say that a function $u : M \to \mathbb{R}$ is (C, α)-Hölder continuous in balls of radius δ if for some constant $C > 0$ we have

$$|u(x) - u(y)| \leqslant C d(x, y)^{\alpha} \quad \text{for all } y \in B(x, \delta).$$

Denote by $|u|_{\alpha,\delta}$ the smallest Hölder constant of u in balls of radius $\delta > 0$.

Let $\delta > 0$ and consider for each $k > 0$ the convex cone of locally Hölder continuous observables defined on M:

$$\mathcal{C}_{k,\delta} = \Big\{ u : u > 0 \text{ and } |u|_{\alpha,\delta} \leqslant k \cdot \inf_{x \in M} u(x) \Big\}. \tag{5.1}$$

Considering the classes of cones of locally Hölder continuous observables, it is possible to provide a more explicit expression to its projective metric.

Lemma 5.4. *Given $\delta > 0$ and $k > 0$ denote the projective metric of $\mathcal{C}_{k,\delta}$ by $\Theta_{k,\delta}$. We have*

$$\Theta_{k,\delta}(u, v) = \log \left(\frac{B_k(u, v)}{A_k(u, v)} \right),$$

where

$$A_{k,\delta}(u, v) := \inf_{d(x,y) < \delta, z \in M} \frac{k d(x, y)^{\alpha} v(z) - (v(x) - v(y))}{k d(x, y)^{\alpha} u(z) - (u(x) - u(y))}$$

and

$$B_{k,\delta}(u, v) := \sup_{d(x,y) < \delta, z \in M} \frac{k d(x, y)^{\alpha} v(z) - (v(x) - v(y))}{k d(x, y)^{\alpha} u(z) - (u(x) - u(y))}.$$

Proposition 5.5. *Given $k > 0$, $\delta > 0$, then for each $u \in \mathcal{C}_{k,\delta}$, we have*

$$\sup_{z \in M} u(z) \leqslant \inf_{z \in M} u(z)(1 + m \cdot k \cdot \operatorname{diam}(M)^{\alpha}).$$

Proof. By $u \in \mathcal{C}_{\hat{\lambda} k,\delta}$, we have $|u|_{\alpha,\delta} \leqslant \hat{\lambda} \cdot k \cdot \inf_{x \in M} u(x)$. Since M is compact, there exist y and z with $u(y) = \sup_{x \in M} u(x)$ and $u(z) = \inf_{x \in M} u(x)$. Assume that $\{B(x_i, \delta/3)\}_{i=1}^{m-1}$ is a family of balls that covers M. Then we define a graph $G = (V, E)$ by $V = \{x_i : 1 \leqslant i \leqslant m - 1\}$ and $E = \{(x_i, x_k) : 1 \leqslant i, k \leqslant m - 1 \text{ and } d(x_i, x_k) < \delta\}$. It follows from the connectedness of M that G is a connected graph. Hence, there exists an integer $2 \leqslant s \leqslant m + 1$ and a sequence $\{p_k\}_{k=1}^{s}$ of points such that $p_1 = y$, $p_s = z$, and $d(p_k, p_{k+1}) < \delta$ for each integer $1 \leqslant k \leqslant s - 1$. Thus,

$$|u(y) - u(z)| \leqslant \sum_{k=1}^{s-1} |u(p_{k+1}) - u(p_k)| \leqslant \sum_{k=1}^{s-1} |u|_{\alpha,\delta} \cdot d(p_k, p_{k+1})^{\alpha} \leqslant (s - 1)|u|_{\alpha,\delta} \cdot [\operatorname{diam}(M)]^{\alpha}.$$

Hence, we obtain that

$$\sup_{x \in M} u(x) \leqslant \inf_{x \in M} u(x) + (s - 1)|u|_{\alpha,\delta} \cdot [\operatorname{diam}(M)]^{\alpha} \leqslant \inf_{x \in M} u(x)(1 + m \cdot \hat{\lambda} \cdot k \cdot [\operatorname{diam}(M)]^{\alpha}),$$

which proves the claim. $\square$

This lemma immediately follows [?, Lemma 4.2]. Then we show that a cone $\mathcal{C}_{\hat{\lambda}k,\delta}$ has a finite $\Theta_{k,\delta}$-diameter.

Proposition 5.6. *Given $k > 0$, $\delta > 0$, and $0 < \hat{\lambda} < 1$. Assume that there exists a family of $m-1$ balls of radius $\delta/3$ that covers M. Then we have $\mathcal{C}_{\hat{\lambda}k,\delta} \subseteq \mathcal{C}_{k,\delta}$ and*

$$\operatorname{diam}_{\Theta_{k,\delta}}(\mathcal{C}_{\hat{\lambda}k,\delta}) \leqslant \Delta(\hat{\lambda}, k, \delta), \tag{5.2}$$

where $\Delta(\hat{\lambda}, k, \delta) := 2\log\left(\frac{1+\hat{\lambda}}{1+\hat{\lambda}}\right) + 2\log(1 + m \cdot \hat{\lambda} \cdot k \cdot \operatorname{diam}(M)^\alpha)$.

Proof. Let $u, v \in \mathcal{C}_{\hat{\lambda}k,\delta}$. Then by Lemma 5.4 one can obtain the following estimate

$$\Theta_{k,\delta}(u,v) \leqslant \log\left(\frac{k \cdot \sup_{x\in M} v(x) + \hat{\lambda}\cdot k \cdot \inf_{x\in M} v(x)}{k \cdot \inf_{x\in M} v(x) - \hat{\lambda}\cdot k \cdot \inf_{x\in M} v(x)} \cdot \frac{k \cdot \sup_{x\in M} u(x) + \hat{\lambda}\cdot k \cdot \inf_{x\in M} u(x)}{k \cdot \inf_{x\in M} u(x) - \hat{\lambda}\cdot k \cdot \inf_{x\in M} u(x)}\right).$$

Using the Claim, we have

$$\begin{aligned}
\Theta_{k,\delta}(u,v) &\leqslant \log\left(\frac{(1+m\cdot\hat{\lambda}\cdot k\,[\operatorname{diam}(M)]^\alpha)(1+\hat{\lambda})\inf_{x\in M} u(x)}{(1-\hat{\lambda})\inf_{x\in M} u(x)}\right) \\
&+ \log\left(\frac{(1+m\cdot\hat{\lambda}\cdot k\,[\operatorname{diam}(M)]^\alpha)(1+\hat{\lambda})\inf_{x\in M} v(x)}{(1-\hat{\lambda})\inf_{x\in M} v(x)}\right) \\
&\leqslant 2\log\left(\frac{1+\hat{\lambda}}{1-\hat{\lambda}}\right) + 2\log\left(1+m\cdot\hat{\lambda}\cdot k\cdot[\operatorname{diam}(M)]^\alpha\right).
\end{aligned}$$

So far we have shown that $\mathcal{C}_{\hat{\lambda}k,\delta} \subseteq \mathcal{C}_{k,\delta}$ and $\operatorname{diam}_{\Theta_{k,\delta}}(\mathcal{C}_{\hat{\lambda}k,\delta}) \leqslant \Delta(\hat{\lambda}, k, \delta)$. □

In our setting a cone of locally Hölder continuous functions is sent by the transfer operator into a cone with finite diameter.

Theorem 5.7. *Let*

$$\delta = \left(\frac{(\frac{1}{\sqrt{e^{-c}}}-1)\kappa}{C_0}\right)^{\frac{1}{\alpha_0}} \text{ and } \hat{\lambda} = e^{\varepsilon_\phi}\left(\frac{(\deg(f)-q)\sigma^{-\alpha} + qL^\alpha[1+(L-1)]^\alpha}{\deg(f)}\right) + \varepsilon_\phi 2mL^\alpha \operatorname{diam}(M)^\alpha.$$

Then under Assumptions 1 and 2, $\mathcal{L}(\mathcal{C}_{k,\delta}) \subset \mathcal{C}_{\hat{\lambda}k,\delta}$ for each $k \geqslant (m(\operatorname{diam}(M))^\alpha)^{-1}$.

Proof. Let $k \geqslant (m(\operatorname{diam}(M))^\alpha)^{-1}$ and take $u \in \mathcal{C}_{k,\delta}$. Note that

$$\mathcal{L}(u)(x) = \sum_{y\in f^{-1}(x)} u(y)\exp(\phi(y)) \geqslant \deg(f)\cdot \inf_{x\in M} u(x) \cdot \exp\bigl(\inf_{x\in M}\phi(x)\bigr).$$

Considering $x, y \in M$ with $d(x,y) < \delta$. For each integer $1 \leqslant i \leqslant \deg(f)$, let x_i and y_i be the preimages of x and y, respectively. We obtain that

$$\begin{aligned}
\frac{|\mathcal{L}(u)|_{\alpha,\delta}}{\inf_{z\in M}(\mathcal{L}(u)(z))} &\leqslant \frac{|\mathcal{L}(u)(x)-\mathcal{L}(u)(y)|}{d(x,y)^\alpha \cdot \inf_{z\in M}\mathcal{L}(u)(z)} \\
&\leqslant \frac{\sum_{i=1}^{\deg(f)} |u(x_i)\exp(\phi(x_i)) - u(y_i)\exp(\phi(y_i))|}{\deg(f)\cdot d(x,y)^\alpha \cdot \inf_{z\in M} u(z)\cdot \exp(\inf_{z\in M}\phi(z))} \\
&\leqslant \frac{\sum_{i=1}^{\deg(f)} |u(x_i)-u(y_i)||\exp(\phi(x_i))| + \sum_{i=1}^{\deg(f)} |u(y_i)||\exp(\phi(x_i)) - \exp(\phi(y_i))|}{\deg(f)\cdot d(x,y)^\alpha \cdot \inf_{z\in M} u(z)\cdot \exp(\inf_{z\in M}\phi(z))} \\
&\leqslant k\frac{(\deg(f)-q)\cdot\sigma^{-\alpha} + q\cdot 2^{1-\alpha}L^\alpha}{\deg(f)} e^{\epsilon_\phi} + 2mL^\alpha diam(M)^\alpha k \frac{|\exp(\phi)|_\alpha}{\exp(\inf_{z\in M}\phi(z))}.
\end{aligned}$$

We are now able to define explicitly the constant $\hat{\lambda}$ as follows.

$$\hat{\lambda} = \frac{(\deg(f)-q)\cdot\sigma^{-\alpha} + q\cdot 2^{1-\alpha}L^\alpha}{\deg(f)} e^{\varepsilon_\phi} + 2mL^\alpha \mathrm{diam}(M)^\alpha \varepsilon_\phi. \tag{5.3}$$

□

5.4. **Exponential convergence of the transfer operator to its eigenfunction.** Let $\lambda > 0$ be the spectral radius of the transfer operator $\mathcal{L}$. Consider the normalized operador $\tilde{\mathcal{L}} = \lambda^{-1}\mathcal{L}$.

Theorem 5.8. *Let $R_1 = 1 + mk diam(M)^\alpha$. Let h be an eigenfunction of the transfer operator: $\mathcal{L}h = \lambda h$. Given an observable $\psi \in \mathcal{C}_{k,\delta}$ with $\int \psi \, d\nu = 1$ we have*

$$\|\tilde{\mathcal{L}}^n(\psi) - h\| \leqslant 3R_1\Delta(1-e^{\Delta})^n \quad \text{for every } n \in \mathbb{N}.$$

Proof. Let $\varphi \in \mathcal{C}_{k,\delta}$ with $\int \varphi \, d\nu = 1$. Since ν is the reference measure associated to λ and $\mu = h\nu$ we have for every $j \geqslant 1$

$$\int \lambda^{-j}\mathcal{L}^j_\phi(\varphi)\, d\nu = \int \lambda^{-j}\varphi \, d(\mathcal{L}^j_\phi)^*\nu = \int \varphi \, d\nu = 1 = \int h \, d\nu.$$

Thus, for every $j \geqslant 1$ we derive

$$\inf \frac{\lambda^{-j}\mathcal{L}^j_\phi(\varphi)}{\lambda^{-j}\mathcal{L}^j(h)} = \inf\frac{\lambda^{-j}\mathcal{L}^j(\varphi)}{h} \leqslant 1 \leqslant \sup\frac{\lambda^{-j}\mathcal{L}^j(\varphi)}{h} = \sup\frac{\lambda^{-j}\mathcal{L}^j(\varphi)}{\lambda^{-j}\mathcal{L}^j(h)}.$$

Moreover

$$\begin{aligned}
e^{-\Delta\tau^j} \leqslant A_k(\tilde{\mathcal{L}}^j_{\phi_*}(\varphi), \tilde{\mathcal{L}}^j_{\phi_*}(h)) &\leqslant \inf\frac{\tilde{\mathcal{L}}^j_{\phi_*}(\varphi)}{h} \\
&\leqslant 1 \\
&\leqslant \sup\frac{\tilde{\mathcal{L}}^j_{\phi_*}(\varphi)}{h} \leqslant B_k(\tilde{\mathcal{L}}^j_{\phi_*}(\varphi), \tilde{\mathcal{L}}^j_{\phi_*}(h)) \leqslant e^{\Delta\tau^j}.
\end{aligned} \tag{5.4}$$

Thus for all $j \geqslant 1$, we have:

$$\left|\tilde{\mathcal{L}}^j_{\phi_*}(\varphi) - h\right|_0 \leqslant |h|_0 \left|\frac{\tilde{\mathcal{L}}^j_{\phi_*}(\varphi)}{h} - 1\right|_0 \leqslant |h|_0\left(e^{\Delta\tau^j} - 1\right) \leqslant L_1\tau^j.$$

Moreover, the inequality (5.4) also gives us

$$e^{-\Delta\tau^j} \leqslant A_k(\tilde{\mathcal{L}}^j_{\phi_*}(\varphi), \tilde{\mathcal{L}}^j_{\phi_*}(h)) \leqslant \frac{kd(x,y)^\alpha \tilde{\mathcal{L}}^j_{\phi_*}(\varphi)(z) - \Big(\tilde{\mathcal{L}}^j_{\phi_*}(\varphi)(x) - \tilde{\mathcal{L}}^j_{\phi_*}(\varphi)(y)\Big)}{kd(x,y)^\alpha h(z) - (h(x) - h(y))},$$

and

$$\frac{kd(x,y)^\alpha \tilde{\mathcal{L}}^j_{\phi_*}(\varphi)(z) - \Big(\tilde{\mathcal{L}}^j_{\phi_*}(\varphi)(x) - \tilde{\mathcal{L}}^j_{\phi_*}(\varphi)(y)\Big)}{kd(x,y)^\alpha h(z) - (h(x) - h(y))} \leqslant B_k(\tilde{\mathcal{L}}^j_{\phi_*}(\varphi), \tilde{\mathcal{L}}^j_{\phi_*}(h)) \leqslant e^{\Delta\tau^j}.$$

Therefore for every $j \geqslant 1$ we obtain

$$\begin{aligned}
&\Big|\tilde{\mathcal{L}}^j_{\phi_*}(\varphi) - h\Big|_\alpha = \sup_{x\neq y} \frac{\Big(\tilde{\mathcal{L}}^j_{\phi_*}(\varphi)(y) - h(y)\Big) - \Big(\tilde{\mathcal{L}}^j_{\phi_*}(\varphi)(x) - h(x)\Big)}{d(x,y)^\alpha} \\
&\leqslant \Big|\tilde{\mathcal{L}}^j_{\phi_*}(\varphi) - h\Big|_0 + \\
&+ \left|\frac{kd(x,y)^\alpha \tilde{\mathcal{L}}^j_{\phi_*}(\varphi)(z) - \Big(\tilde{\mathcal{L}}^j_{\phi_*}(\varphi)(x) - \tilde{\mathcal{L}}^j_{\phi_*}(\varphi)(y)\Big)}{kd(x,y)^\alpha h(z) - (h(y) - h(x))} - 1\right| \cdot \left|kh(z) - \frac{h(y) - h(x)}{d(x,y)^\alpha}\right| \\
&\leqslant L_1\tau^j + (e^{\Delta\tau^j} - 1)\cdot(k\,|h|_0 + |h|_\alpha) \leqslant L_2\tau^j.
\end{aligned}$$

Thus for every $j \geqslant 1$ we have the inequality

$$\Big\|\tilde{\mathcal{L}}^j_{\phi_*}(\varphi) - h\Big\| = \Big|\tilde{\mathcal{L}}^j_{\phi_*}(\varphi) - h\Big|_0 + \Big|\tilde{\mathcal{L}}^j_{\phi_*}(\varphi) - h\Big|_{\alpha,\delta} \leqslant L_3\tau^j.$$

Now, given $n \geqslant 1$ write $n = 3j + r$ with $j < n$ and $0 \leqslant r < 3$. Since $\mathcal{L}_{\phi_*}$ is a bounded operator and $\mathcal{L}_{\phi_*}h = \lambda h$, we conclude that

$$\begin{aligned}
\big\|\lambda^{-n}\mathcal{L}^n_{\phi_*}(\varphi) - h\big\| &= \Big\|\lambda^{-r}\mathcal{L}^r_{\phi_*}\Big(\lambda^{-3j}\mathcal{L}^{3j}_{\phi_*} - h\Big)\Big\| \\
&\leqslant \big\|\lambda^{-1}\mathcal{L}_{\phi_*}\big\|^r \cdot \Big\|\tilde{\mathcal{L}}^j_{\phi_*}(\varphi) - h\Big\| \leqslant L\tau^n.
\end{aligned}$$

□

5.5. **Jacobians and conformal measures.**

Definition 5.9. Let η be a probability measure on M. We say that a function $J_\eta f$ is a *Jacobian* of η with respect to f if it satisfies:

$$\eta\,(f\,(A)) = \int_A J_\eta f\, d\eta,$$

for all mensurable sets A such that $f|_A$ is injective.

In other words, given the measure η, $f_*\eta(B) = \eta\big(f(B)\big)$ for all measurable subsets $B \in M$.

The Radon-Nikodym theorem ensures that the Jacobian is essentially unique. Further, it coincides with the Radon-Nikodym derivative of the measure $f_*\eta$ with respect to η, $J_\eta f = \frac{df_*\eta}{d\eta}$.

5.6. **Computability of the equilibrium state.** In this section we prove the computability of the equilibrium state μ associated to the map f and the potential φ satisfying Assumptions 1 and 2. The strategy is to prove the computability of the Jacobian function and apply Theorem 4.7 in [**?**].

Theorem 5.10. *There exists an algorithm satisfying the following property:*

For each $x_0 \in X$, each $n \in \mathbb{N}$, and (applicable system), the algorithm outputs a rational 2^{-n}-approximation for the value of $J_\varphi(x_0)$, where the function $J_\varphi\colon X \to \mathbb{R}$ is defined by

$$J_\varphi(x) := \frac{u_\varphi(T(x))}{u_\varphi(x)} \exp(P(T,\varphi) - \varphi(x)) \quad \textit{for each } x \in X, \tag{5.5}$$

on the following input

(i) two algorithms computing φ and T, respectively......

Before the proof of this theorem, we shall do some preparations. We begin with designing an algorithm that computes the function $\mathcal{L}_\varphi(\mathbb{1})$.

Proposition 5.11. *Let $(X, \rho, \mathcal{S})$ be a computable metric space, X a recursively compact set, and $T\colon X \to X$ a computable function. Assume that T is a local homeomorphism on X, then the inverse T^{-1} of T is computable.*

Proof. □

As an immediate consequence of Proposition 5.11 and the computability of the exponential function, one gets the computability of the Ruelle–Perron–Frobenius operator in the following sense:

Corollary 5.12. *There exists an algorithm satisfying the following property:*

For each $x_0 \in X$, each n, $m \in \mathbb{N}$, and (applicable system), this algorithm outputs a rational 2^{-n}-approximation for the value of $\mathcal{L}_\varphi^m(\mathbb{1})(x_0)$, after inputting an oracle of the point x_0, the constants n, $m \in \mathbb{N}$, and data (i) through (iv) from Theorem 5.10 in this algorithm.

Now we establish the computability of the value of the pressure $P(T, \varphi)$.

Lemma 5.13. *There exists an algorithm satisfying the following property:*

For each $n \in \mathbb{N}$ and (applicable system), this algorithm outputs a rational 2^{-n}-approximation for the topological pressure $P(T, \varphi)$, after inputting the constant $n \in \mathbb{N}$ and data (i) through (iv) from Theorem 5.10 in this algorithm.

Proof. □

Then we can show that the function $u_\varphi\colon X \to \mathbb{R}$ is computable.

Lemma 5.14. *There exists an algorithm that satisfies the following property:*

For each $x_0 \in X$, each $n \in \mathbb{N}$, and (applicable system), this algorithm outputs a rational 2^{-n}-approximation for the value of $u_\varphi(x_0)$, after inputting an oracle of the point x_0, the constant $n \in \mathbb{N}$, and data (i) through (iv) from Theorem 5.10 in this algorithm.

Proof. □

We are now ready to establish Theorem 5.10.

Proof of Theorem 5.10. By Corollary 5.12, Lemmas 5.13, and 5.14, it follows from (5.5) that the function J_φ is computable. □

Theorem 5.15. *Let $(X, \rho, \mathcal{S})$ be a computable metric space, X be a recursively compact set, $\phi\colon X \to \mathbb{R}$ a computable function, and $T\colon X \to X$ be a computable map with finite topological entropy, finitely many singular points, and the property that the number of preimages of each point is uniformly bounded. Suppose that U is a lower semi-computable open subset of X satisfying that there exists a sequence $\{U_j\}_{j\in\mathbb{N}}$ of uniformly lower semi-computable open subsets of X that satisfy the following properties:*

(a) *$U = \bigcup_{j\in\mathbb{N}} U_j$.*

(b) *T is injective and open on U_j for each $j \in \mathbb{N}$.*

(c) *$\{T(U_j)\}_{j\in\mathbb{N}}$ is a sequence of lower semi-computable open sets.*

Suppose that $J\colon X \to (0, +\infty)$ is a Borel function that is upper semi-computable on U and bounded on U. Then $\mathcal{M}(X, T; U, J)$ is a recursively compact subset of $\mathcal{P}(X)$.

Assume, in addition, that $C \subseteq X$ is a recursively compact set with $C \smallsetminus B \subseteq U$ for some subset $B \subseteq X$, and J satisfies the following properties:

(d) *There exists a Borel function* $h\colon X \to \mathbb{R}$ *such that for each* $x \in C \smallsetminus B$,

$$J(x) = \exp(P(T,\phi) - \phi(x) + h(T(x)) - h(x)).$$

(e) $\sum\limits_{y\in T^{-1}(x)\cap U} \frac{1}{J(y)} = 1$ *for each* $x \in \bigcap_{i\in\mathbb{N}_0} T^i(U)$.

Then the following statements are true:

(i) $\mathcal{M}(X,T;U,J) \cap \mathcal{P}(X;C\smallsetminus B) = \mathcal{E}(T,\phi)\cap\mathcal{P}(X;C\smallsetminus B)$.

(ii) *If* $\mathcal{M}(X,T;U,J)\cap\mathcal{P}(X;C)\cap\mathcal{K} = \mathcal{E}(T,\phi)\cap\mathcal{P}(X;C\smallsetminus B) = \{\mu_\phi\}$ *for some recursively compact set* $\mathcal{K}\subseteq\mathcal{P}(X)$, *then the equilibrium state* μ_ϕ *is a computable measure.*

Consider the above result in the case where $B=\emptyset$, $C=X$, and T is a homeomorphism. Note that a local homeomorphism is always an open map which satisfies that the number of preimages of each point is uniformly bounded. Then we have the following corollary.

Theorem 5.16. *Let* $(X,\,\rho,\,\mathcal{S})$ *be a computable metric space,* X *be a recursively compact set,* $\phi\colon X\to\mathbb{R}$ *a computable function, and* $T\colon X\to X$ *be a computable local homeomorphism with finite topological entropy. Suppose that there exists a sequence* $\{U_k\}_{k\in\mathbb{N}}$ *of uniformly lower semi-computable open subsets of* X *that satisfy the following properties:*

(a) $X = \bigcup_{k\in\mathbb{N}} U_k$.

(b) T *is injective on* U_k *for each* $j\in\mathbb{N}$.

(c) $\{T(U_k)\}_{k\in\mathbb{N}}$ *is a sequence of lower semi-computable open sets.*

Suppose that $J\colon X\to(0,+\infty)$ *is a Borel function that satisfies the following properties:*

(d) J *is upper semi-computable on* X *and bounded on* X.

(e) *There exists a continuous function* $h\colon X\to\mathbb{R}$ *such that for each* $x\in X$,

$$J(x) = \exp(P(T,\phi) - \phi(x) + h(T(x)) - h(x)).$$

(f) $\sum\limits_{y\in T^{-1}(x)} \frac{1}{J(y)} = 1$ *for each* $x\in\bigcap_{i\in\mathbb{N}_0} T^i(X)$.

If $\mathcal{E}(T,\phi) = \{\mu_\phi\}$, *then the equilibrium state* μ_ϕ *is a computable measure.*

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